

Sales Analysis performance Report

1. Introduction

This project focuses on analyzing sales performance using Superstore sales data. The objective of this analysis is to understand sales trends, regional performance, category performance, and product performance. The analysis helps businesses make better decisions and improve overall sales strategies.

2. Dataset Overview

The dataset contains sales-related information including:

- Order ID
- Order Date
- Customer Name
- Region
- State
- Category
- Product Name
- Sales

The dataset was analyzed using Python libraries such as Pandas and Matplotlib in Google Colab.

3. Data Cleaning

The following data cleaning steps were performed:

- Removed duplicate records
- Checked missing values
- Filled missing values
- Converted date columns into datetime format
- Verified dataset consistency

These steps improved the quality of the dataset for further analysis.

4. Feature Engineering

New features were created to improve analysis.

Order Month

The order month was extracted from the order date column to analyze monthly sales trends and seasonality.

5. KPI Metrics

The following KPI metrics were analyzed:

Total Sales

Calculated the total revenue generated from sales.

Average Sales

Calculated the average sales value.

Highest Sales Product

Identified the product generating the highest sales.

Highest Sales Region

Identified the region contributing the highest sales.

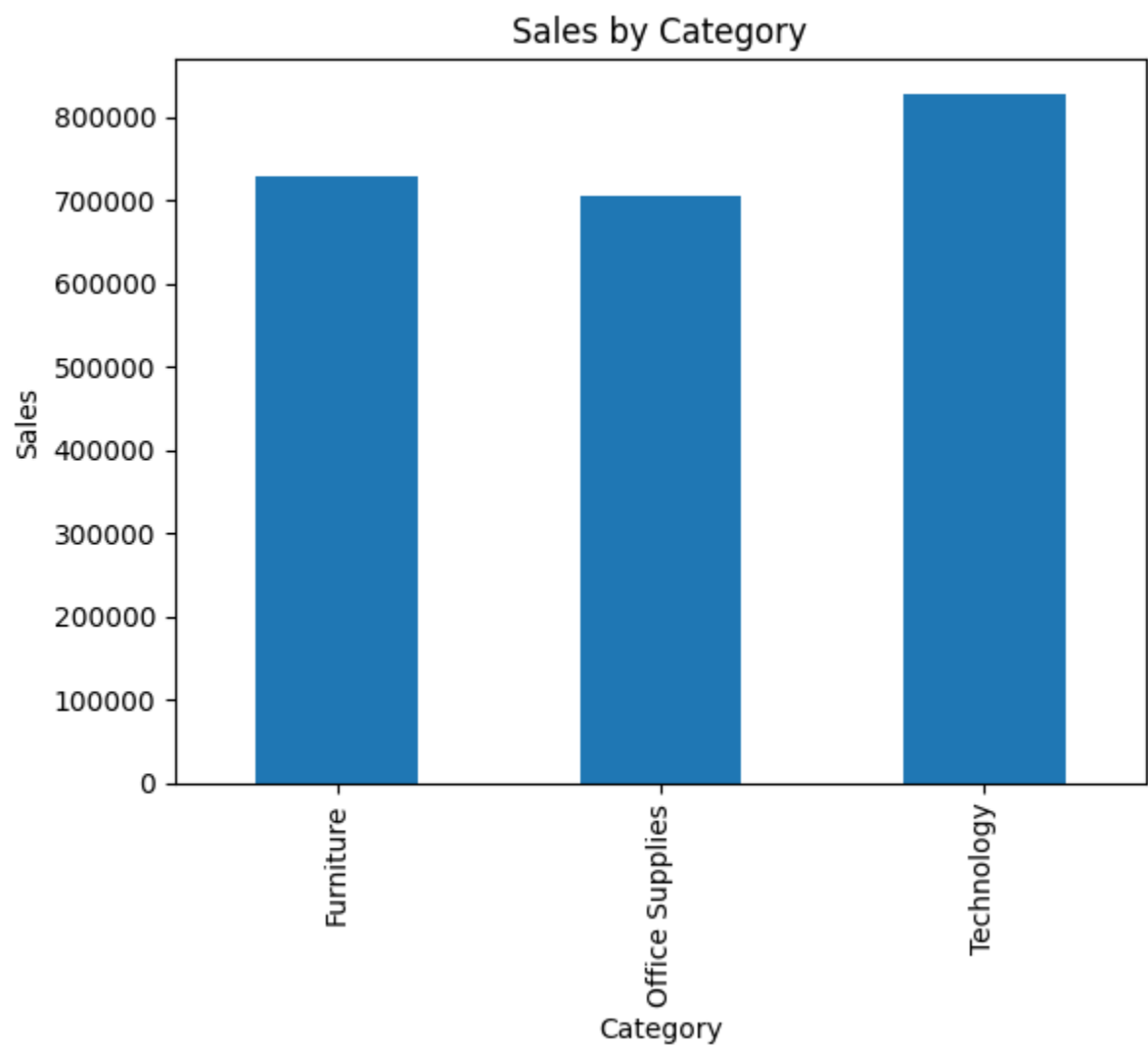
Highest Sales State

Identified the state generating the highest revenue.

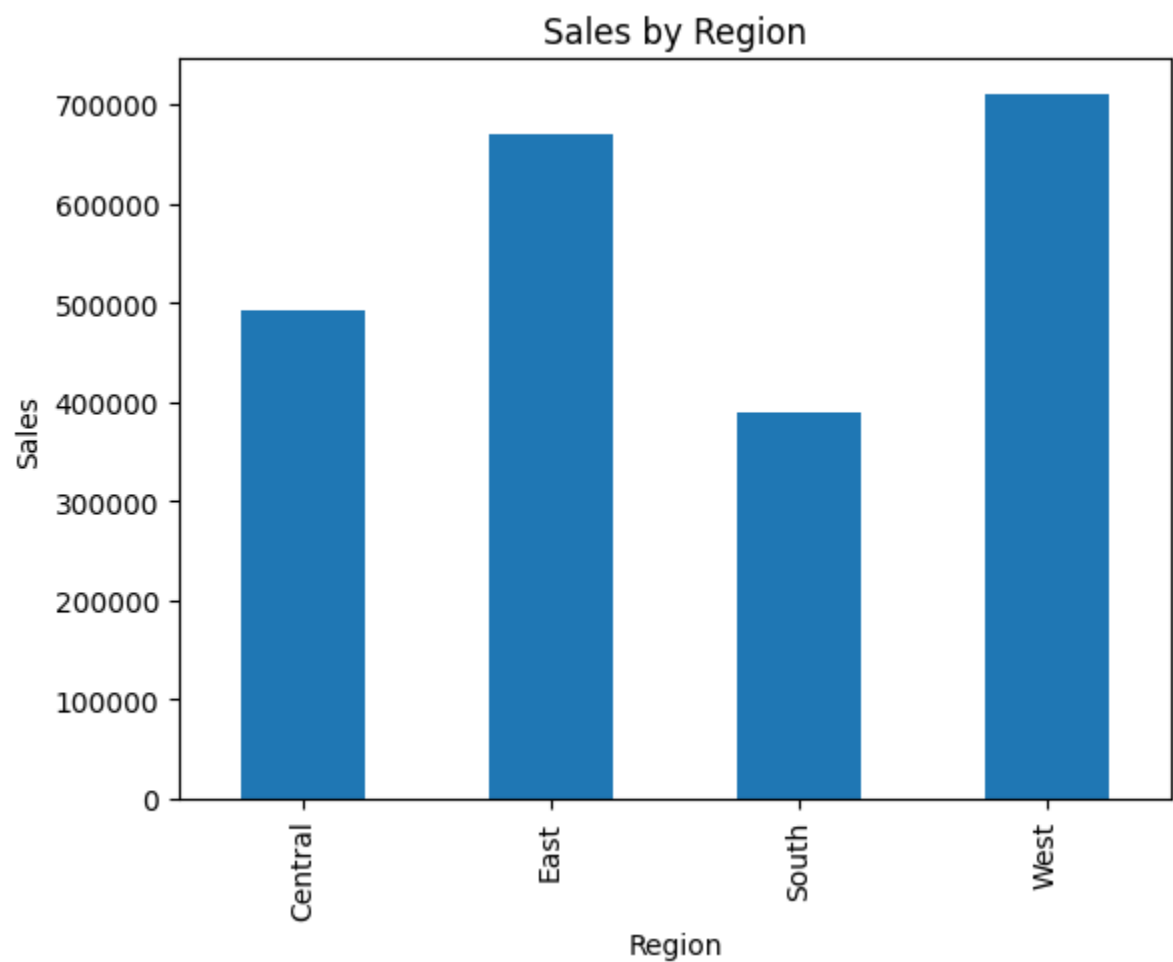
6. Visualizations

The following visualizations were created:

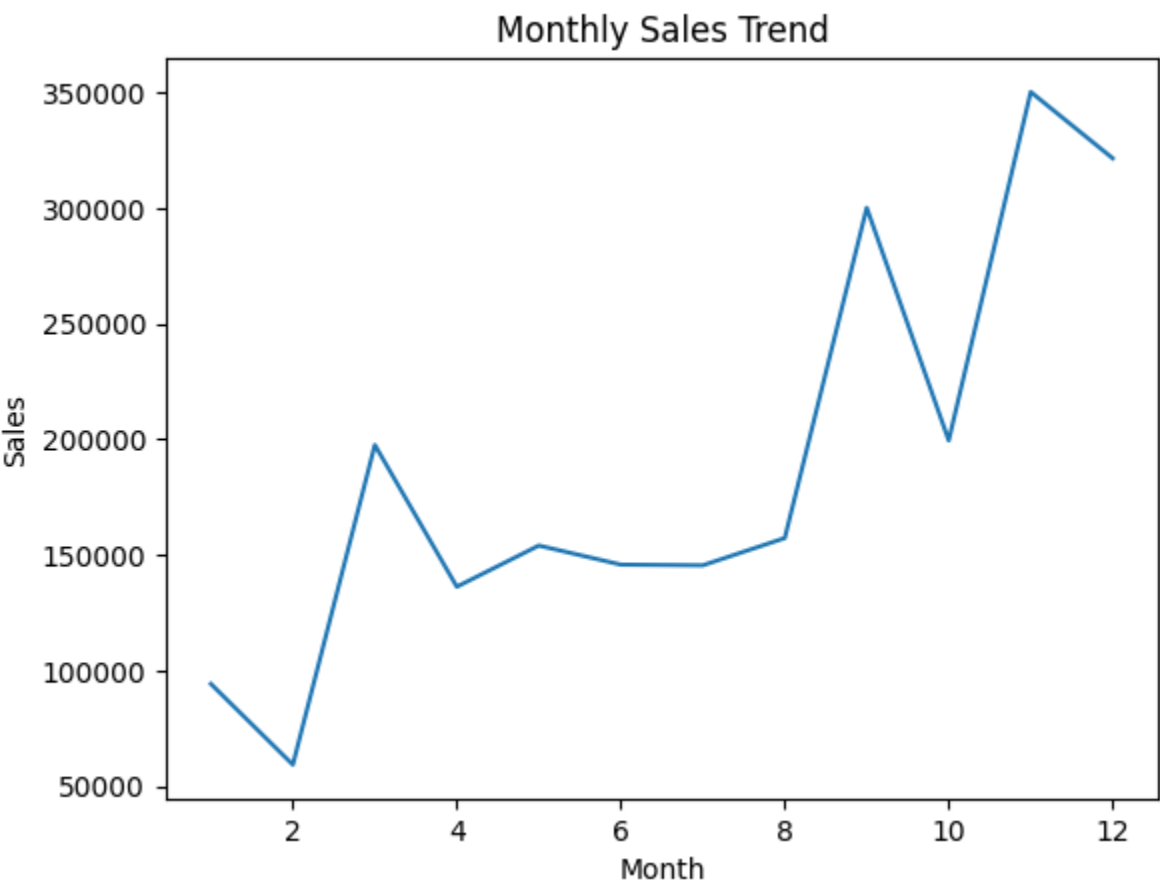
Sales by Category



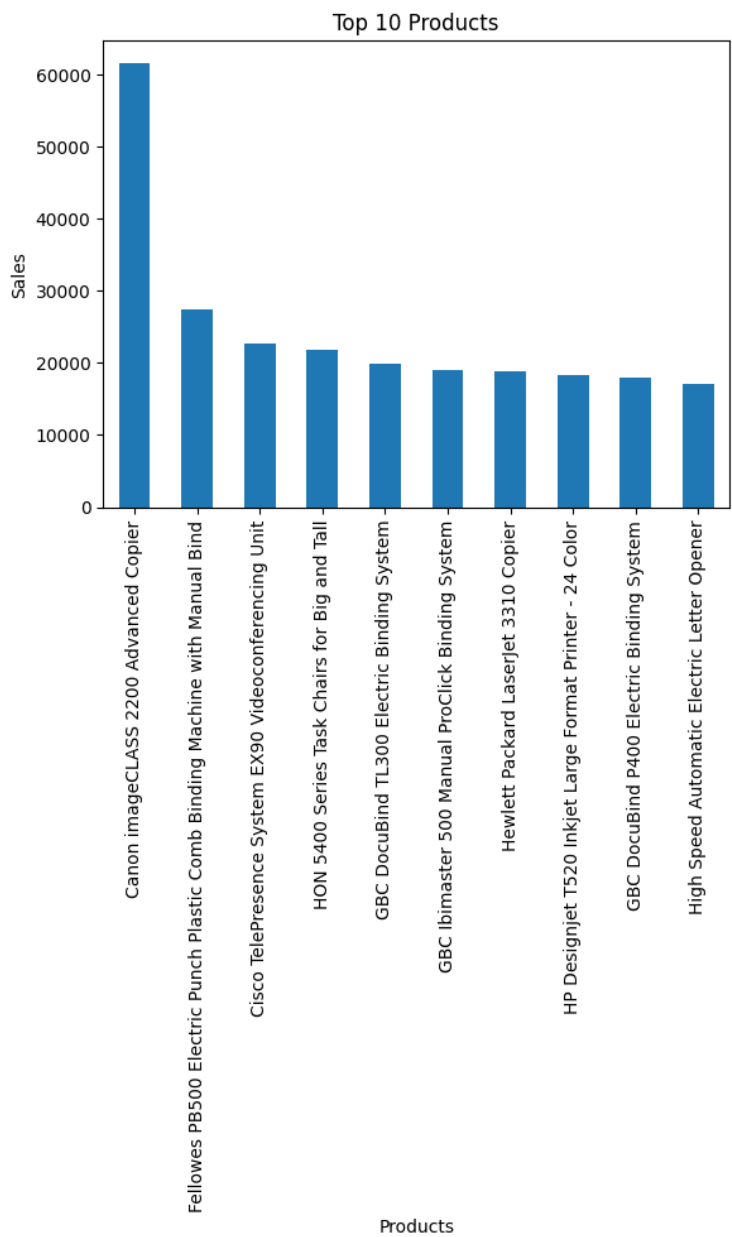
Sales by Region



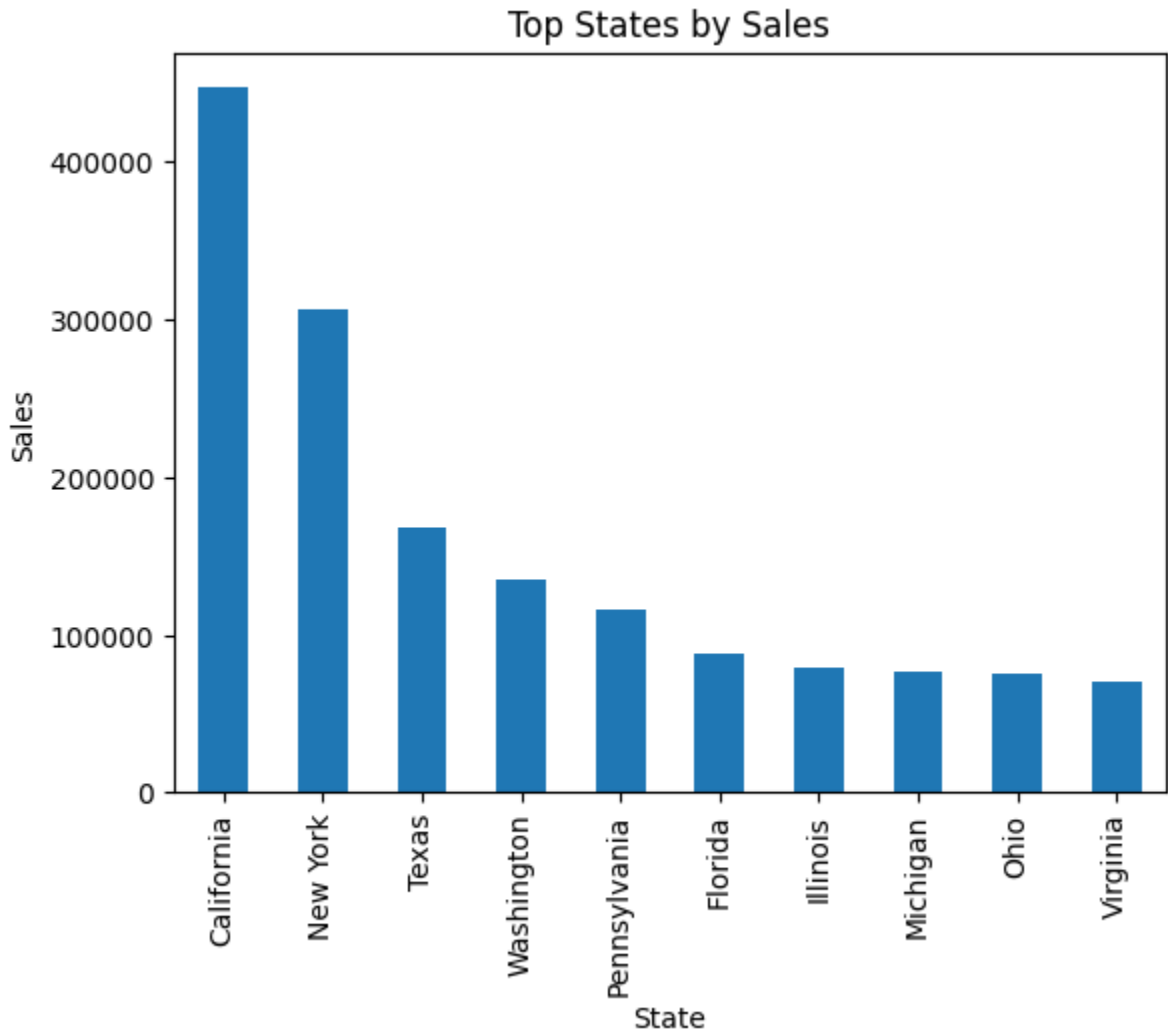
Monthly Sales Trend



Top 10 Products



Top States by Sales



These visualizations helped in understanding sales distribution and business performance trends.

7. Findings / Observations

- Technology category generated high sales.
- Some regions performed better than others.
- Monthly sales showed changing trends throughout the year.
- Certain products contributed major revenue.

- Some states generated significantly higher sales than others.

8. Recommendations

1. Focus on marketing strategies in high-performing regions.
2. Promote top-selling products to increase revenue.
3. Improve sales performance in low-performing areas.
4. Use seasonal campaigns to boost monthly sales.
5. Increase inventory for products with high demand.

9. Conclusion

The sales performance analysis provided valuable insights into sales trends, product performance, and regional sales distribution. The analysis can help businesses improve marketing strategies, inventory planning, and overall business performance.